

Louis Fengqi Hu

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Authorized to work in the U.S. (F-1 OPT/CPT — no sponsorship required for internship/OPT period)

SUMMARY

Full-stack and AI engineer with hands-on experience building production-grade LLM-powered applications. Proficient in Java, Spring Boot, React, Redis, RAG pipelines, and LLM API integration (OpenAI, Gemini). Academic background in ML and computer vision combined with end-to-end project delivery spanning system design, implementation, and deployment.

EDUCATION

Johns Hopkins University *Master of Engineering in Computer Science, Aug. 2025 – Dec. 2026 (Expected)*

Coursework: Machine Learning, Computer Vision, Software System Design, Artificial Intelligence

Zhejiang Gongshang University *Bachelor of Engineering, Software Engineering, Sep. 2021 – Jun. 2025*

Coursework: Data Structures, Operating Systems, Software Testing, Design Patterns, Databases, Data Analysis

PROFESSIONAL SKILLS

Languages: Java, Python, JavaScript, TypeScript, C, SQL, PHP

Frameworks & Databases: Spring Boot, React, Vue 3, Node.js, Express, PostgreSQL, MongoDB, Redis, Elasticsearch

DevOps/Tools: Docker, Nginx, Git, RabbitMQ, JUnit, Vitest, Jenkins, Vercel, Firebase, MySQL

AI / ML: PyTorch, OpenCV, Pandas, NumPy, OpenAI API, Gemini API, RAG, YOLOv8

PUBLICATIONS

Python's Evolution on Stack Overflow: An Empirical Analysis of Topic Trends *Aug. 2023 – Aug. 2024*

First Author, Journal of Computer Languages (COLA), Vol. 84, Elsevier, 2025

- Analyzed 2M+ Python-related Stack Overflow posts, classified posts into 5 hotness grades using a multi-rater annotation to uncover topic trends and shifts in library usage.
- Trained and optimized a GBDT model for post-hotness classification via grid search and eigenvalue amplification, achieving 91.38% accuracy (all posts) and 90.56% (unanswered posts), outperforming SVM, KNN, RF, and GNB baselines.

PROFESSIONAL EXPERIENCE

Hundsun Technologies Inc. *Oct. 2024 – Dec. 2024*

Software Engineer (Intern), Hangzhou, China

- Contributed to backend development of transaction middleware services using Java and Spring Boot, participating in feature implementation and code review cycles. Integrated internal LLM tooling to automate code review annotations and made a standard prompt document for coding generation, boosting coding efficiency by an estimated 30%.
- Collaborated with senior engineers to refactor legacy modules, reducing complexity across 3 core service components and resolved 5+ business logic defects, reducing error rates in order-processing flows.

PROJECTS

WearWise - AI-Powered Personal Wardrobe *Jan. 2026 – Present*

- Built and deployed a full-stack AI wardrobe platform using React, Express, and MongoDB on Vercel, enabling users to manage clothing inventories and receive AI-generated outfit recommendations.
- Integrated Gemini APIs with RAG to build a context-aware recommendation engine that factors in user preferences, wardrobe inventory, and real-time weather. Built virtual try-on images using NanoBanana 2, allowing users to visualize clothing items.
- Implemented asynchronous image upload processing via RabbitMQ, reducing upload latency and decoupling the upload pipeline from the recommendation service.

Hybrid defect detection and validation framework for ceramic tiles *Sep. 2025 – Dec. 2025*

- Designed an automated surface defect detection system using YOLOv8 for real-time ceramic tile quality inspection.
- Improved model architecture by integrating Coordinate Attention and SPD-Conv modules, achieving a 28% reduction in model size vs. the YOLOv8s baseline while maintaining inference speed and detection accuracy on the test set.

Kitty Gourmet Food Hall - Online Shopping Platform *Oct. 2023 – Jun. 2024*

- Led end-to-end development of a full-stack shopping platform supporting user management, product catalogue, shopping cart, and order processing for a food retail client. Managed the full project lifecycle including sprint planning, feature implementation, server maintenance, and cross-stage technical issue resolution.
- Built with Vue 3 and Spring Boot, containerized with Docker, and deployed via Nginx; Redis caching for frequently accessed product data enabled the system to handle 1,000+ concurrent requests with response times under 500 ms.

Programmer Forum Recommendation System Based on User Profiling *Sep. 2023 – Mar. 2024*

Software Copyright Owner (Jointly)

- Built a content recommendation engine with Node.js and Vue 3 that dynamically surfaces forum posts based on real-time user interaction signals, reducing information overload for active users.
- Modelled user behavior using Jaccard similarity for interest profiling; applied a hotness-based cold-start strategy for new users.